

量子统计

2021. 11. 12

微扰论

解 $\hat{H}|\psi\rangle = E|\psi\rangle$ 的方法除微扰论外, 还有绝热近似, 变分法, 半经典近似

一级微扰公式: $\hat{H} = \hat{H}_0 + \hat{H}'$

可解 小的修正 ($\hat{H}' \ll \hat{H}_0 = \hat{H}^{(0)}$)

$$\hat{H}|\psi\rangle = (\hat{H}_0 + \hat{H}')|\psi\rangle = (E_n^{(0)} + E_n^{(1)})|\psi\rangle$$

$E_n = E_n^{(0)} + E_n^{(1)} + E_n^{(2)} + \dots$ 其中 $E_n^{(n)}$ 和 $\psi^{(n)}$ 与 $\hat{H}'$ 的 n 次方成正比

$$\psi_n = \psi_n^{(0)} + \psi_n^{(1)} + \psi_n^{(2)} + \dots$$

$$(\hat{H}_0 + \hat{H}')(\psi_n^{(0)} + \psi_n^{(1)} + \psi_n^{(2)} + \dots) = (E_n^{(0)} + E_n^{(1)} + E_n^{(2)} + \dots)(\psi_n^{(0)} + \psi_n^{(1)} + \psi_n^{(2)} + \dots)$$

零级方程 $\hat{H}_0 \psi_n^{(0)} = E_n^{(0)} \psi_n^{(0)}$

← 必须最完全

$$\hat{H}_0 \psi_n^{(1)} + \hat{H}' \psi_n^{(0)} = E_n^{(0)} \psi_n^{(1)} + E_n^{(1)} \psi_n^{(0)}$$

一级方程 $(\hat{H}_0 - E_n^{(0)}) \psi_n^{(1)} = -(\hat{H}' - E_n^{(1)}) \psi_n^{(0)}$

$$\hat{H}_0 \psi_n^{(2)} + \hat{H}' \psi_n^{(1)} = E_n^{(0)} \psi_n^{(2)} + E_n^{(1)} \psi_n^{(1)} + E_n^{(2)} \psi_n^{(0)}$$

二级方程 $(\hat{H}_0 - E_n^{(0)}) \psi_n^{(2)} = -(\hat{H}' - E_n^{(1)}) \psi_n^{(1)} + E_n^{(2)} \psi_n^{(0)}$

一级
二级
求解

据 $\hat{H}_0$ 是 Hermitic 算符, $\{\psi_n^{(0)}\}$ 构成完备系

$$\text{把 } \psi_n^{(1)} \text{ 向 } \{\psi_m^{(0)}\} \text{ 展开, } \psi_n^{(1)} = \sum_m a_{nm}^{(1)} \psi_m^{(0)}$$

$$\sum_m a_{nm}^{(1)} (\hat{H}_0 - E_n^{(0)}) \psi_m^{(0)} = - \sum_m a_{nm}^{(1)} (\hat{H}' - E_n^{(1)}) \psi_m^{(0)}$$

$$\sum_m a_{nm}^{(1)} (E_m^{(0)} - E_n^{(0)}) \psi_m^{(0)} = - \sum_m a_{nm}^{(1)} (\hat{H}' - E_n^{(1)}) \psi_m^{(0)}$$

取

$$\sum_m a_{nm}^{(1)} (\hat{H}_0 - E_n^{(0)}) \psi_m^{(0)} = - \sum_m a_{nm}^{(1)} (\hat{H}' - E_n^{(1)}) \psi_m^{(0)}$$

$$\Rightarrow \sum_m a_{nm}^{(1)} (E_m^{(0)} - E_n^{(0)}) \psi_m^{(0)} = - \sum_m a_{nm}^{(1)} (\hat{H}' - E_n^{(1)}) \psi_m^{(0)}$$

$$\Rightarrow a_{nk}^{(1)} (E_k^{(0)} - E_n^{(0)}) = - \sum_m a_{nm}^{(1)} \langle \psi_k^{(0)} | \hat{H}' | \psi_m^{(0)} \rangle + E_n^{(1)} \delta_{kn}$$

$$= - \langle \psi_k^{(0)} | \hat{H}' | \psi_n^{(0)} \rangle + E_n^{(1)} \delta_{kn}$$

$$= -H_{kn}' + E_n^{(1)} \delta_{kn}$$

$$\Rightarrow 0 = -H_{kn}' + E_k^{(1)} \Rightarrow E_k^{(1)} = H_{kk}'$$

$$\text{若 } n \neq k, \quad a_{nk} = \frac{-H_{kn}'}{E_k^{(0)} - E_n^{(0)}} = \frac{H_{kn}'}{E_n^{(0)} - E_k^{(0)}}, \quad \psi_n^{(1)} = \sum_{k \neq n} \frac{H_{kn}'}{E_n^{(0)} - E_k^{(0)}} \psi_k^{(0)}$$

二級微擾能 $(\hat{H}_0 - E_n^{(0)}) \psi_n^{(1)} = -(\hat{H}' - E_n^{(1)}) \psi_n^{(0)} + E_n^{(2)} \psi_n^{(0)}$

$$\sum_{m \neq n} a_{nm}^{(1)} (\hat{H}_0 - E_n^{(0)}) \psi_m^{(0)} = -\sum_{m \neq n} a_{nm}^{(1)} (\hat{H}' - E_n^{(1)}) \psi_m^{(0)} + E_n^{(2)} \psi_n^{(0)}$$

$$\sum_{m \neq n} a_{nm}^{(1)} (E_m^{(0)} - E_n^{(0)}) \psi_m^{(0)} = -\sum_{m \neq n} a_{nm}^{(1)} \frac{H_{mn}'}{E_n^{(0)} - E_m^{(0)}} \hat{H}' \psi_m^{(0)} + E_n^{(1)} \sum_{m \neq n} a_{nm}^{(1)} \frac{H_{mn}'}{E_n^{(0)} - E_m^{(0)}} \psi_m^{(0)} + E_n^{(2)} \psi_n^{(0)}$$

$$a_{nn}^{(1)} (E_n^{(0)} - E_n^{(0)}) = -\sum_{m \neq n} \frac{H_{mn}'}{E_n^{(0)} - E_m^{(0)}} H_{em}' + E_n^{(1)} \sum_{m \neq n} \frac{H_{mn}'}{E_n^{(0)} - E_m^{(0)}} +$$

$$0 = -\sum_{m \neq n} \frac{H_{mn}' H_{nm}'}{E_n^{(0)} - E_m^{(0)}} + E_n^{(1)}$$

$$= -\sum_{m \neq n} \frac{|H_{mn}'|^2}{E_n^{(0)} - E_m^{(0)}} + E_n^{(1)} \Rightarrow E_n^{(1)} = \sum_{m \neq n} \frac{|H_{mn}'|^2}{E_n^{(0)} - E_m^{(0)}}$$

$$E_n = E_n^{(0)} + \sum_{m \neq n} \frac{|H_{mn}'|^2}{E_n^{(0)} - E_m^{(0)}}$$

$$\psi_n^{(1)} = \sum_{m \neq n} \frac{H_{mn}'}{E_n^{(0)} - E_m^{(0)}} \psi_m^{(0)}, \text{ 微擾論適用條件 } \left| \frac{H_{mn}'}{E_n^{(0)} - E_m^{(0)}} \right| \ll 1 \quad (n \neq m)$$

例題：[基本模型無限深方阱，一維諧振子，中心力場]

1. 各向同性的非極性分子電介質在外電場作用下極化，計算感生電矩

正離子在平衡位置附近振動 ($\hat{H}_0$)

$$\left(-\frac{\hbar^2}{2\mu} \frac{\partial^2}{\partial x^2} + \frac{1}{2} \mu \omega^2 x^2\right) = \hat{H}_0, \text{ 記一維諧振子的波函數是 } |n\rangle,$$

$$E_n = (n + \frac{1}{2}) \hbar \omega, \quad n = 0, 1, 2, \dots$$

加了電場後 (E 沿 x 正方向), $H' = -qEx \Rightarrow \hat{H}' = -qE\hat{x} = -qEx$
($x=0$ 為勢能零點)

$$H_{mn}' = \langle m | H' | n \rangle = -qE \langle m | x | n \rangle$$

$$x |n\rangle = \sqrt{\frac{\hbar}{\mu\omega}} \left(\sqrt{\frac{n+1}{2}} |n+1\rangle + \sqrt{\frac{n}{2}} |n-1\rangle \right) \Rightarrow \langle m | x | n \rangle = \sqrt{\frac{\hbar}{\mu\omega}} \left(\sqrt{\frac{n+1}{2}} \delta_{m,n+1} + \sqrt{\frac{n}{2}} \delta_{m,n-1} \right)$$

$$H_{mn}' = -qE \sqrt{\frac{\hbar}{\mu\omega}} \left(\sqrt{\frac{n+1}{2}} \delta_{m,n+1} + \sqrt{\frac{n}{2}} \delta_{m,n-1} \right)$$

$$H_{nn}' = -qE \langle n | x | n \rangle = 0$$

$$E_n^{(1)} = \sum_{m \neq n} \frac{|H_{mn}'|^2}{E_n^{(0)} - E_m^{(0)}} = \sum_{m \neq n} \frac{q^2 E^2 \frac{\hbar}{\mu\omega}}{E_n^{(0)} - E_m^{(0)}} \left(\sqrt{\frac{n+1}{2}} \delta_{m,n+1} + \sqrt{\frac{n}{2}} \delta_{m,n-1} \right)^2$$

$$= \frac{q^2 E^2 \hbar}{\mu\omega} \cdot \frac{(n+1) \cdot \frac{\hbar}{2}}{(n+1) \cdot \frac{\hbar}{2} - \frac{n \cdot \hbar}{2}} = \frac{q^2 E^2 \hbar}{\mu\omega} \left(-\frac{\frac{n}{2}}{\frac{\hbar}{2}} + \frac{\frac{n+1}{2}}{-\frac{\hbar}{2}} \right) = \frac{q^2 E^2 \hbar}{\mu\omega} \left(-\frac{n}{\hbar} - \frac{n+1}{\hbar} \right) = -\frac{q^2 E^2}{2\mu\omega^2}$$

一级微扰 $|\psi_n\rangle = |n\rangle + \sum_{m \neq n} \frac{H_{mn}^{(1)}}{E_n^{(0)} - E_m^{(0)}} |m\rangle$

$$= |n\rangle + \frac{qE}{\omega \sqrt{\hbar m \omega}} \left(\sqrt{\frac{n+1}{2}} |n+1\rangle + \sqrt{\frac{n}{2}} |n-1\rangle \right)$$

在外电场 $\langle x \rangle = \langle K | x | K \rangle = 0$

加的电场, $\langle x \rangle = \langle \psi_n | x | \psi_n \rangle = \frac{qE}{\omega \sqrt{\hbar m \omega}} \left(\sqrt{\frac{n+1}{2}} x_{n,n+1} + \sqrt{\frac{n}{2}} x_{n,n-1} \right)$

$$= \frac{qE}{\hbar m \omega^2}$$

电矩 $D = 2 \langle x \rangle |q| = \frac{2q^2 E}{\hbar m \omega^2}$, $\frac{1}{\hbar m \omega^2} K = \frac{D}{E} = \frac{2q^2}{\hbar m \omega^2}$

图

定态微扰论：简并情形。

零级近似： $\hat{H}_0 |\psi_n^{(0)}\rangle = E_n^{(0)} |\psi_n^{(0)}\rangle$

$E_n^{(0)}$ 简并, $\hat{H}_0 |\psi_{ni}^{(0)}\rangle = E_n^{(0)} |\psi_{ni}^{(0)}\rangle, i=1, 2, \dots, k$

令 $\psi_n^{(0)} = \sum_{i=1}^k C_{ni}^{(0)} \psi_{ni}^{(0)}$

一级近似： $(\hat{H}_0 - E_n^{(1)}) \psi_n^{(1)} = -(\hat{H}' - E_n^{(1)}) \psi_n^{(0)}$

$$= - \sum_{i=1}^k C_{ni}^{(0)} (\hat{H}' - E_n^{(1)}) \psi_{ni}^{(0)}$$

$\langle \psi_{nj}^{(0)} | \hat{H}_0 - E_n^{(1)} | \psi_n^{(1)} \rangle = \langle \psi_{nj}^{(0)} | \hat{H}' | \psi_n^{(0)} \rangle - E_n^{(1)} \langle \psi_{nj}^{(0)} | \psi_n^{(0)} \rangle$

$= \sum_{i=1}^k C_{ni}^{(0)} \langle \psi_{nj}^{(0)} | \hat{H}' | \psi_{ni}^{(0)} \rangle - E_n^{(1)} \delta_{ji}$

$0 = - \sum_{i=1}^k C_{ni}^{(0)} (\langle \psi_{nj}^{(0)} | \hat{H}' | \psi_{ni}^{(0)} \rangle - E_n^{(1)} \delta_{ji})$

$\Rightarrow \sum_{i=1}^k C_{ni}^{(0)} (H'_{ji} - E_n^{(1)} \delta_{ji}) = 0$

量子統計

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期末考：三簡答題、(精讀)：統計力學的三大統計 (Fermi, Bosiman, Bose)

Stark 效應

原子或分子在外電場的作用下，有能級或譜線分裂的現象，原子發射的譜線頻率是對稱的，間隔數

$$(E \sim 100 \text{ MV/cm})$$

$$E \propto n^2$$

~~Stark 效應~~

1913年. Stark 發射, 1916年. Einstein 納入量子力學

1917年. Nobel Prize 1926年 Schrödinger 用波動力學解釋

或原子能級裂距 $\propto E$ (一級 Stark)

或金屬原子能級裂距 $\propto E^2$ (二級 Stark)

$$u(r) = -\frac{e^2}{r}, \quad -\vec{D} \cdot \vec{E} = -(-e) \cdot \vec{r} \cdot \vec{E} = erE \cos \theta$$

例：氫原子能級在靜電場的分裂

$$\hat{H}' = - (e\vec{r}) \cdot \vec{E} = erE \cos \theta$$

$$\text{Q. } \hat{H} = \hat{H}_0 + \hat{H}'$$

$$\hat{H}_0 |nlm\rangle = E_n^{(0)} |nlm\rangle, \quad E_n = -\frac{e^2}{2a} \cdot \frac{1}{n^2}, \quad a = \frac{\hbar^2}{\mu e^2} \text{ Bohr radius}$$

(1) 基態能差一級修正

$$E_1^{(1)} = \langle 100 | \hat{H}' | 100 \rangle = 0$$

$$\vec{r}(r, \theta, \varphi) \rightarrow -\vec{r}(r, \pi - \theta, \pi + \varphi)$$

$$Y_{lm}(\pi - \theta, \pi + \varphi) = (-1)^l Y_{lm}(\theta, \varphi)$$

$$\cos \theta Y_{lm} = \sqrt{\frac{(l+1)^2 - m^2}{(2l+1)(2l+3)}} Y_{l+1,m} + \sqrt{\frac{l^2 - m^2}{(2l-1)(2l+1)}} Y_{l-1,m}$$

(2) 第一激發態一級修正

波函數: $|200\rangle, |210\rangle, |211\rangle, |21-1\rangle$

宇稱: $+$ $-$ $-$ $-$

$$[\hat{H}', \hat{L}_z] = [eE r \cos \theta, \hat{L}_z] = 0$$

$$\langle nlm | \hat{H}' \hat{L}_z - \hat{L}_z \hat{H}' | nl'm' \rangle = 0$$

$$\Leftrightarrow (m' - m) \langle nlm | \hat{H}' | nl'm' \rangle = 0$$

$$\Rightarrow \langle nlm | \hat{H}' | nl'm' \rangle = \langle nlm | \hat{H}' | nl'm' \rangle \delta_{m'm}$$

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正常 Zeeman 效應：磁場在外，磁場（磁場） H_{LS} 可忽略

$$\hat{H} = \hat{H}_0 + \frac{\mu_B B_0}{\hbar} (\hat{L}_z + 2\hat{S}_z)$$

採用非耦合表象 $\{\hat{H}, \hat{L}^2, \hat{L}_z, \hat{S}^2, \hat{S}_z\}$ $|nlm_l \frac{1}{2} m_s\rangle = |nlm_l m_s\rangle$

$$E_{nlm_l m_s} = \langle nlm_l m_s | \hat{H}_0 + \frac{\mu_B B_0}{\hbar} (\hat{L}_z + 2\hat{S}_z) | nlm_l m_s \rangle$$

$$= E_{nl}^{(0)} + \frac{\mu_B B_0}{\hbar} (m_l + 2m_s) \cdot \hbar = E_{nl}^{(0)} + \mu_B B_0 (m_l + 2m_s)$$

選擇定則是 $\Delta l = \pm 1, \Delta m_l = -1, 0, +1, \Delta m_s = 0$ 反常 Zeeman 效應： $\hat{H} = \hat{H}_0 + H_{LS} + \hat{W}$

$$= \hat{H}_0 + H_{LS} + \frac{\mu_B B_0}{\hbar} (\hat{J}_z + \hat{S}_z)$$

$$E_{nlj m_j} \approx \langle nlj m_j | \hat{H} | nlj m_j \rangle = E_{nlj}^{(0)} + \mu_B B_0 m_j + \langle nlj m_j | \hat{S}_z | nlj m_j \rangle \frac{\mu_B}{\hbar}$$

$$|nlj m_j\rangle = \sum_{m_l m_s} C_{nlm_l m_s}^{j m_j} |nlm_l m_s\rangle$$

$$\langle nlj m_j | \hat{S}_z | nlj m_j \rangle = \begin{cases} \frac{\hbar}{2j} m_j & l \neq 0, j = l + \frac{1}{2} \\ -\frac{\hbar}{2j} m_j & l \neq 0, j = l - \frac{1}{2} \\ m_j \hbar & l = 0 \end{cases}$$

選擇定則 $\Delta l = \pm 1, \Delta j = 0, \pm 1, \Delta m_j = 0, \pm 1$

量子態隨時間的演化

Hamilton 量隨時間含時體系的量子躍遷的微擾論

$$\hat{H}_0 |n\rangle = E_n |n\rangle$$

$$\text{若 } t=0, |\psi(0)\rangle = |\psi_n\rangle$$

$$t>0, \hat{H} = \hat{H}_0 + \hat{H}'(t)$$

$$\text{以 } \hat{H}_0 \text{ 表象展開, } |\psi(t)\rangle = \sum_n C_n(t) |n\rangle \exp(-i \frac{E_n}{\hbar} t), C_n(0) = \delta_{nn}$$

$$= \sum_n C_n(t) |\psi_n\rangle$$

$$P_{n \leftarrow n} = \left| \langle n | \psi(t) \rangle \right|^2 = |C_n(t)|^2$$

$$i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = \hat{H}_0 + \hat{H}'(t) |\psi(t)\rangle, \text{ 設 } \hat{H}'(t) \text{ 不含 } \frac{\partial}{\partial t} \text{ 項}$$

$$i\hbar \sum_n \left(\dot{C}_n(t) - i \frac{E_n}{\hbar} C_n(t) \right) |n\rangle e^{-i \frac{E_n}{\hbar} t} = \sum_n C_n(t) (\hat{H}_0 + \hat{H}'(t)) |n\rangle e^{-i \frac{E_n}{\hbar} t}$$

$$= \sum_n C_n(t) (E_n + \hat{H}'(t)) |n\rangle e^{-i \frac{E_n}{\hbar} t}$$

$$\Rightarrow i\hbar \sum_n \dot{C}_n(t) |n\rangle e^{-i \frac{E_n}{\hbar} t} = \sum_n C_n(t) \hat{H}'(t) |n\rangle e^{-i \frac{E_n}{\hbar} t}$$

用 $|k\rangle$ 内积

$$\begin{cases} i\hbar \dot{C}_{k'k}(t) e^{-i\frac{E_{k'}}{\hbar}t} = \sum_n C_{kn}(t) \langle k' | \hat{H}(t) | n \rangle e^{-i\frac{E_n}{\hbar}t} \\ C_{k'k}(0) = \delta_{k'k} \end{cases}$$

$$\begin{cases} i\hbar \dot{C}_{k'k} = \sum_n C_{kn} \langle k' | \hat{H}(t) | n \rangle e^{i\omega_{k'n}t} \\ C_{k'k}(0) = \delta_{k'k} \end{cases}$$

$$\dot{C}_{k'k}(t) = C_{k'k}(0) + \frac{1}{i\hbar} \int_0^t \sum_n C_{kn}(t') \langle k' | \hat{H}(t') | n \rangle e^{i\omega_{k'n}t'} dt'$$

$$= \delta_{k'k} + \frac{1}{i\hbar} \int_0^t \sum_n \hat{H}'_{k'n}(t') C_{kn}(t') e^{i\omega_{k'n}t'} dt'$$

$$= \delta_{k'k} + \frac{1}{i\hbar} \int_0^t \sum_n \hat{H}'_{k'n}(t') e^{i\omega_{k'n}t'} \left(\delta_{kn} + \frac{1}{i\hbar} \int_0^{t'} \sum_m \hat{H}'_{nm}(t'') C_{mk}(t'') e^{i\omega_{nm}t''} dt'' \right) dt'$$

$$\approx \delta_{k'k} + \frac{1}{i\hbar} \int_0^t \hat{H}'_{k'k}(t') e^{i\omega_{k'k}t'} dt'$$

量子變統計

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2. (5.1) 在 $r \geq r_0$ 時, $U(r) = -k \frac{Ze^2}{r}$

在 $r \leq r_0$ 時, $U(r) = -k \frac{Ze^2}{2r_0} (3r_0^2 - r^2)$

設 $\hat{U}_0(r) = U_0(r) = -k \frac{Ze^2}{r}$

$$\hat{U}'(r) = U'(r) = \begin{cases} -k \frac{Ze^2}{2r_0} (3r_0^2 - r^2) + k \frac{Ze^2}{r} & r \leq r_0 \\ 0 & r \geq r_0 \end{cases}$$

$$\hat{H}_0 = -\frac{\hbar^2}{2m} \nabla^2 + \hat{U}_0, \quad \hat{H}' = \begin{cases} -k \frac{Ze^2}{2r_0} (3r_0^2 - r^2) + k \frac{Ze^2}{r}, & r \leq r_0 \\ 0, & r \geq r_0 \end{cases}$$

基態波函數 $\psi_{100} = R_{10}(r) Y_{00}(\theta, \varphi) = |100\rangle$, $E_1^{(0)} = \frac{m_e Z^2 e^4}{2\hbar^2}$, $R_{10}(r) = 2 \left(\frac{Z}{a}\right)^{3/2} \exp\left(-\frac{2Zr}{a}\right)$

基態能量的一級修正: $E_1^{(1)} = \langle 100 | \hat{H}' | 100 \rangle = \langle R_{10} | \hat{H}' | R_{10} \rangle$

$$= \int_0^{r_0} R_{10}(r) \cdot k \frac{Ze^2}{2r_0} \left[\frac{1}{r} + \frac{r^2}{2r_0^3} - \frac{3}{2r_0} \right] \cdot R_{10}(r) \cdot r^2 dr$$

$$= k \frac{Ze^2}{2r_0} \cdot 4 \left(\frac{Z}{a}\right)^3 \int_0^{r_0} \left(r + \frac{r^4}{2r_0^3} - \frac{3r^2}{2r_0} \right) \cdot \exp\left(-\frac{2Zr}{a}\right) dr$$

$$\approx k \frac{Ze^2}{2r_0} \cdot 4 \left(\frac{Z}{a}\right)^3 \int_0^{r_0} \left(r + \frac{r^4}{2r_0^3} - \frac{3r^2}{2r_0} \right) dr$$

$$= k \frac{Ze^2}{2r_0} \cdot 4 \left(\frac{Z}{a}\right)^3 \cdot \left(\frac{1}{2} r_0^2 + \frac{1}{10} r_0^2 - \frac{1}{2} r_0^2 \right)$$

$$e_s = e\sqrt{k}$$

$$= \frac{2Z^4 e^2 r_0^2}{5a^3}, \quad \text{所以 } E_0 = \frac{m_e Z^2 e^4}{2\hbar^2} + \frac{2Z^4 e^2 r_0^2}{5a^3}$$

3. (5.2) $\hat{H}_0 = \frac{\hat{L}^2}{2I}$, $\hat{H}' = -\vec{D} \cdot \vec{E} = -DE \cos\theta$ (不妨設 $\vec{E}$ 沿 z 軸方向)

基態波函數 $Y_{00}(\theta, \varphi) = |00\rangle = \frac{1}{\sqrt{4\pi}}$, $E_0^{(0)} = 0 \times (0+1) \hbar^2 / 2I = 0$

$$\hat{H}'_{00} = \langle Y_{00} | \hat{H}' | Y_{00} \rangle = \langle 00 | \hat{H}' | 00 \rangle$$

$$= \langle 00 | \hat{H}' | 00 \rangle$$

$$= -DE \langle 00 | \cos\theta | 00 \rangle$$

$$= -DE$$

$$E_0^{(1)} = \langle 00 | \hat{H}' | 00 \rangle$$

$$= -DE \langle 00 | \cos\theta | 00 \rangle$$

$$= -DE \int_0^\pi \int_0^{2\pi} \frac{1}{\sqrt{4\pi}} \cos\theta \cdot \frac{1}{\sqrt{4\pi}} \sin\theta d\varphi d\theta$$

$$= -DE \cdot \frac{1}{2} \int_0^\pi \cos\theta \sin\theta d\theta = 0$$

$$E_0^{(1)} = \sum_{l \neq 0} \sum_{m=0}^l \frac{|\langle lm | \hat{H}' | 00 \rangle|^2}{-l(l+1) \frac{\hbar^2}{2I}}$$

$$= \sum_{l=1}^\infty \sum_{m=-l}^l \frac{D^2 E^2 \left| \frac{1}{\sqrt{3}} \langle lm | 10 \rangle \right|^2}{-l(l+1) \cdot \frac{\hbar^2}{2I}}$$

$$= \sum_{l=1}^\infty \sum_{m=-l}^l \frac{\frac{D^2 E^2}{3} \delta_{lm} \delta_{m0}}{-l(l+1) \cdot \frac{\hbar^2}{2I}}$$

$$= -\frac{D^2 E^2 \hbar^2}{3\hbar^2}$$

$$E_0 = -\frac{D^2 E^2 \hbar^2}{3\hbar^2}$$

4. (3.3) $E_1^{(0)} = E_{01}, E_2^{(0)} = E_{02}$

對應本征函數分別為 $|1\rangle, |2\rangle$

$$E_1^{(1)} = \langle 1 | \hat{H}' | 1 \rangle = H'_{11} = b, E_2^{(1)} = \langle 2 | \hat{H}' | 2 \rangle = H'_{22} = b$$

$$E_1^{(2)} = \sum_{m \neq 1} \frac{1}{E_1^{(0)} - E_m^{(0)}} |\langle m | \hat{H}' | 1 \rangle|^2 = \frac{|\langle 2 | \hat{H}' | 1 \rangle|^2}{E_1^{(0)} - E_2^{(0)}} = \frac{|H_{21}|^2}{E_{01} - E_{02}} = \frac{a^2}{E_{01} - E_{02}}$$

$$E_2^{(2)} = \frac{|\langle 1 | \hat{H}' | 2 \rangle|^2}{E_2^{(0)} - E_1^{(0)}} = \frac{|H_{12}|^2}{E_{02} - E_{01}} = \frac{a^2}{E_{02} - E_{01}}$$

所以 $E_1 = E_{01} + b + \frac{a^2}{E_{01} - E_{02}}, E_2 = E_{02} + b + \frac{a^2}{E_{02} - E_{01}}$

5. (5.5) 基態波函數 $|100\rangle$

終態波函數 $|21-1\rangle$ 或 $|210\rangle$ 或 $|211\rangle$

~~$\hat{H}'(t) = \mathcal{E} = E_0 \exp(-i\omega t) u(t)$~~

不妨設 $\mathcal{E}$ 方向平行於 z 軸. $\hat{H}'(t) = \begin{cases} 0, & t \leq 0 \\ e E_0 e^{-\frac{t}{\tau}} = e E_0 e^{-\frac{t}{\tau}} r \cos \theta u(t) & t > 0 \\ e E_0 e^{-\frac{t}{\tau}} z = -E_0 r e^{-\frac{t}{\tau}} \cos \theta, & t > 0 \end{cases} = e E(t) r \cos \theta$

~~$\langle 21-1 | \hat{H}'(t) | 100 \rangle$~~

$$\langle 21m | \hat{H}'(t) | 100 \rangle = e E(t) \langle 21m | r \cos \theta | 100 \rangle = \langle R_{21} | r | R_{10} \rangle \langle 1m | \cos \theta | 00 \rangle e E(t)$$

$$= \langle R_{21} | r | R_{10} \rangle \cdot \frac{1}{\sqrt{3}} \langle 1m | 10 \rangle e E(t)$$

$$= \langle R_{21} | r | R_{10} \rangle \cdot \frac{1}{\sqrt{3}} \delta_{m0} e E(t)$$

~~$$R_{21}(r) = \left(\frac{r}{2a}\right)^{\frac{3}{2}} \left(2 - \frac{r}{a}\right) \exp\left(-\frac{r}{2a}\right)$$~~

~~$$R_{10}(r) = \left(\frac{r}{a}\right)^{\frac{1}{2}} \cdot 2 \exp\left(-\frac{r}{a}\right)$$~~

~~$$\langle R_{21} | r | R_{10} \rangle = \int_0^\infty \left(\frac{r^2}{2a^3}\right)^{\frac{3}{2}} \cdot 2 \left(2 - \frac{r}{a}\right) \cdot \exp\left(-\frac{3r}{2a}\right) \cdot r^2 dr$$~~

=

$$R_{21}(r) = \left(\frac{r}{2a_0}\right)^{\frac{3}{2}} \cdot \frac{r}{a\sqrt{3}} \exp\left(-\frac{r}{2a}\right)$$

$$R_{10}(r) = \left(\frac{r}{a}\right)^{\frac{1}{2}} \cdot 2 \exp\left(-\frac{r}{a}\right)$$

$$\langle R_{21} | r | R_{10} \rangle = \int_0^r \left(\frac{r^2}{2a^3}\right)^{\frac{3}{2}} \cdot 2 \cdot \frac{r}{a\sqrt{3}} \exp\left(-\frac{3r}{2a}\right) r^4 dr$$

$$= \frac{r^3}{2\sqrt{3}a^3} \cdot 2 \cdot \frac{r}{a\sqrt{3}} \cdot \left(\frac{2a}{3r}\right)^4 \Gamma(5)$$

$$= \frac{16\sqrt{2}a}{24\sqrt{3}r} \cdot 4! = \frac{128\sqrt{2}a}{81\sqrt{3}r} = \frac{128\sqrt{2}a}{81\sqrt{3}}$$

$$\langle 21m | \hat{H}'(t) | 100 \rangle = \frac{128\sqrt{2}a}{81\sqrt{3}} \cdot \frac{1}{\sqrt{3}} \delta_{m0} eE(t)$$

$$= \frac{128\sqrt{2}a}{243} \delta_{m0} \cdot eE(t)$$

$$\int_0^t \langle 21m | \hat{H}'(t') | 100 \rangle e^{i\omega_{21}t'} dt'$$

$$= \frac{128\sqrt{2}a}{243} \delta_{m0} e \int_0^t E_0 \exp\left(-\frac{t'}{\tau} + i\omega_{21}t'\right) dt'$$

$$= \frac{128\sqrt{2}a}{243} \delta_{m0} e \cdot \frac{E_0}{\frac{1}{\tau} - i\omega_{21}} \left[1 - \exp\left(-\frac{t}{\tau} + i\omega_{21}t\right) \right] \cdot \frac{1}{\frac{1}{\tau} - i\omega_{21}}$$

$$= \frac{128\sqrt{2}a}{243} \delta_{m0} \cdot e \cdot E_0 \cdot \left[1 - \exp\left(-\frac{t}{\tau} + i\omega_{21}t\right) \right] \cdot \frac{1}{\frac{1}{\tau} - i\omega_{21}}$$

$$P_{21m \leftarrow 100} = \frac{1}{\hbar^2} \left(\frac{128\sqrt{2}a}{243} e E_0 \right)^2 \frac{|1 - \exp(-\frac{t}{\tau} + i\omega_{21}t)|^2}{(\frac{1}{\tau})^2 + \omega_{21}^2} \delta_{m0}$$

$$P_{211 \leftarrow 100} = P_{21-1 \leftarrow 100} = 0$$

$$P_{210 \leftarrow 100} = \frac{1}{\hbar^2} \left(\frac{128\sqrt{2}a}{243} e E_0 \right)^2 \frac{|1 - \exp(-\frac{t}{\tau} + i\omega_{21}t)|^2}{(\frac{1}{\tau})^2 + \omega_{21}^2}$$

$$E_n = \frac{m e^2 E_0^2}{2 \hbar^2 n^2} \quad |100\rangle \rightarrow |210\rangle$$

当 $t \rightarrow \infty$ 时, $P_{211 \leftarrow 100} = P_{21-1 \leftarrow 100} = 0$

$$P_{210 \leftarrow 100} = \frac{1}{\hbar^2} \left(\frac{128\sqrt{2}a}{243} e E_0 \right)^2 \cdot \frac{1}{(\frac{1}{\tau})^2 + \omega_{21}^2}$$

$$\omega_{21} = \frac{E_1 - E_2}{\hbar}$$

6. (补充题1)

(1) 利用分离变量法可得, 能量本征函数是 $|n_1, n_2\rangle = \sqrt{\frac{2}{a}} \sin\left(\frac{n_1 x}{a}\right) \cdot \sqrt{\frac{2}{a}} \sin\left(\frac{n_2 y}{a}\right) |100\rangle \times \dots$

记 $|n_1, n_2\rangle = \sqrt{\frac{2}{a}} \sin\left(\frac{n_1 x}{a}\right) \cdot \sqrt{\frac{2}{a}} \sin\left(\frac{n_2 y}{a}\right)$ 则本征函数可简写 能量 $\frac{\pi^2 \hbar^2}{8ma^2} (n_1^2 + n_2^2)$, $n_1, n_2 \in \mathbb{N}^+$

~~(2) 画出最低能级分别对应态 $|10\rangle$ 和 $|01\rangle$ 的波函数 $E_1^{(0)} = \frac{\pi^2 \hbar^2}{8ma^2}$~~

~~$|11\rangle$~~

(2) x, y 方向的最低能级对应的波函数 $|11\rangle$, $E_1^{(0)} = \frac{\pi^2 \hbar^2}{8ma^2} \times (1^2 + 1^2) = \frac{\pi^2 \hbar^2}{4ma^2}$

$$E_1^{(0)} = \langle 11 | \hat{H}' | 11 \rangle = \lambda \langle 11 | xy | 11 \rangle = \lambda \left(\frac{a}{2}\right)^2$$

$$E_1 = E_1^{(0)} + E_1^{(1)} = \frac{\pi^2 \hbar^2}{4ma^2} + \lambda \left(\frac{a}{2}\right)^2$$

7. (补充题 2)

$$\hat{H}_0 = -\frac{\hbar^2}{2\mu} \nabla^2 + \frac{1}{2} \mu \omega^2 r^2, \quad \hat{H}' = -\lambda (xy + yz + zx)$$

记一维线性谐振子的波函数是 $|n\rangle_x = \left(\frac{\alpha}{\sqrt{\pi} 2^n n!} \right)^{\frac{1}{2}} \exp(-\frac{1}{2} \alpha^2 x^2) H_n(\alpha x)$, $\alpha = \sqrt{\frac{\mu\omega}{\hbar}}$
 $E_n = (n + \frac{1}{2}) \hbar \omega, \quad n \in \mathbb{N}$

则三维各向同性谐振子的波函数是 $|n_1, n_2, n_3\rangle = |n_1\rangle_x |n_2\rangle_y |n_3\rangle_z$, $n_1, n_2, n_3 \in \mathbb{N}$, $E_{n_1, n_2, n_3} = (n_1 + n_2 + n_3 + \frac{3}{2}) \hbar \omega$
 基态波函数 $|000\rangle$, $E_0^{(0)} = \frac{3}{2} \hbar \omega$

$$\begin{aligned} E_0^{(1)} &= \langle 000 | \hat{H}' | 000 \rangle = -\lambda \langle 000 | \sum_{ij} x_i y_j | 000 \rangle \\ &= -\lambda \sum_{ij} \langle 000 | x_i y_j | 000 \rangle \\ &= -\lambda \sum_{ij} \langle 0 | x_i | 0 \rangle_x \langle 0 | y_j | 0 \rangle_y \\ &= 0 \end{aligned}$$

第一级态: $|001\rangle, |010\rangle, |100\rangle$

记: $|1\rangle, |2\rangle, |3\rangle$

$$H'_{nn} = \langle n | \hat{H}' | n \rangle = -\lambda \sum_{ij} \langle n | x_i y_j | n \rangle = 0$$

$$H'_{12} = \langle 001 | \hat{H}' | 010 \rangle$$

$$= -\lambda \langle 001 | xy + yz + zx | 010 \rangle$$

$$= -\lambda \langle 01 | yz | 10 \rangle_{yz}$$

$$= -\lambda \sqrt{\frac{\hbar}{2\mu\omega}} (1) \times \sqrt{\frac{\hbar}{2\mu\omega}} \cdot 1 = -\lambda \cdot \frac{\hbar}{2\mu\omega}$$

$$\Rightarrow H'_{21} = H'_{12} = -\lambda \cdot \frac{\hbar}{2\mu\omega}$$

$$H'_{23} = \langle 010 | \hat{H}' | 100 \rangle$$

$$= -\lambda \langle 01 | xy | 10 \rangle_{xy} = -\lambda \cdot \frac{\hbar}{2\mu\omega} = H'_{12}$$

$$H'_{31} = H'_{11} = -\lambda \cdot \frac{\hbar}{2\mu\omega}, \quad \text{记 } A = \frac{\lambda \hbar}{2\mu\omega}$$

$$\begin{aligned} \text{久期方程} \quad & \begin{vmatrix} -E_1^{(1)} - A & -A & -A \\ -A & -E_1^{(1)} & -A \\ -A & -A & -E_1^{(1)} \end{vmatrix} = (E_1^{(1)} + 2A) \begin{vmatrix} -(E_1^{(1)} - A) & E_1^{(1)} - A & 0 \\ -A & -E_1^{(1)} & -A \\ -A & -A & -E_1^{(1)} \end{vmatrix} \\ & = \begin{vmatrix} -(E_1^{(1)} - A) & 0 & 0 \\ -A & -(A + E_1^{(1)}) & -A \\ -A & -2A & -E_1^{(1)} \end{vmatrix} = -(E_1^{(1)} - A) (E_1^{(1)2} + A E_1^{(1)} - 2A^2) = (E_1^{(1)} - A)^2 (E_1^{(1)} + 2A) = 0 \\ & \Rightarrow E_{1,1}^{(1)} = A (= \text{重根}), \quad E_{1,2}^{(1)} = -2A = -\frac{\lambda \hbar}{\mu\omega} \\ & = \lambda \cdot \frac{\hbar}{2\mu\omega} \end{aligned}$$